Neighbour Embeddings & Contrastive Learning

From Noise-Contrastive Estimation to UMAP

Damrich, Böhm, Hamprecht & Kobak (2023) ICLR arXiv:2206.01816

Code: github.com/berenslab/contrastive-ne

Data Science & Machine Learning
Constructor University Bremen

Lecture 8 — Dimensionality Reduction & Embeddings

Outline

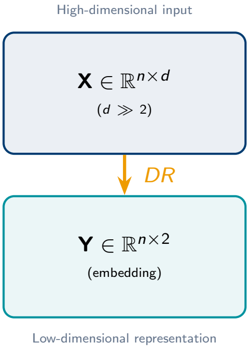
Reminder: the High-Dimensional Data Problem

Fields generating massive high-D data:

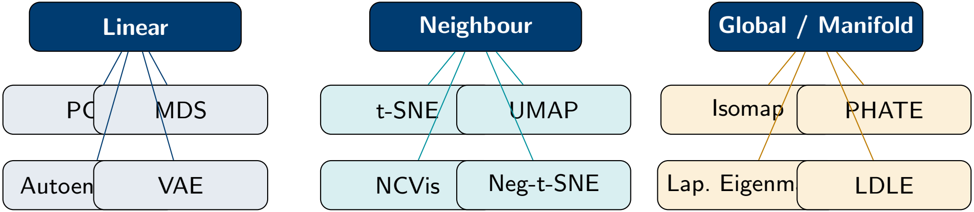
- **Biology:** scRNA-seq — $n \sim 10^6$ cells, $d \sim 30{,}000$ genes
- **Genomics:** SNP arrays — millions of features per sample
- **Neuroscience:** calcium imaging — thousands of neurons $\times$ time
- **NLP/AI:** LLM activations, embedding spaces

Core challenge: We cannot visualise or reason about $d > 3$ dimensions. Yet scientific meaning is **latent in the geometry**.

Damrich et al. (2023) arXiv:2206.01816 · ICLR



A Taxonomy of Embedding Methods



Neighbour methods: de facto standard for visualisation; focus of this lecture

Linear methods: fastest; interpretable dims (PCA as preprocessing)

Global methods: better long-range structure (PHATE, Isomap)

Neighbour Embeddings: The Shared Pipeline

All neighbour embedding methods share a common four-step pipeline:

1. Build k-NN graph → 2. High-dim similarities p_{ij} → 3. Initialise embedding $\mathbf{Y}$ → 4. Optimise loss $\mathcal{L}$

Euclidean distance

Gaussian kernel

PCA or random

Attr. + Repr. forces

t-SNE: symmetric KL divergence

$$\mathcal{L}_{\text{t-SNE}} = \mathrm{KL}(P \| Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

$q_{ij} \propto (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}$ (Cauchy / Student-t kernel)

Damrich et al. (2023) arXiv:2206.01816 — Section 3 & 6

UMAP: binary cross-entropy + NEG

$$\mathcal{L}_{\text{UMAP}} = -\mathbb{E}_{ij \sim p} \log q_{ij} - m\,\mathbb{E}_{ij \sim \xi} \log(1 - q_{ij})$$

Uses **negative sampling**; m pairs per edge

Formal Definitions I: Similarity Kernels and Distributions

High-dimensional similarities. For each point i, find Gaussian bandwidth σ_i via perplexity:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2/2\sigma_i^2)}, \qquad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} \quad \text{(symmetrised)}$$

Low-dimensional kernel (Cauchy / Student-t). t-SNE uses a heavy-tailed kernel to alleviate the crowding problem:

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k\neq l}(1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}} \qquad \text{(t-SNE)}$$

Perplexity. Entropy-based effective neighbourhood size:

$$\mathrm{Perp}(P_i) = 2^{H(P_i)}, \qquad H(P_i) = -\sum_j p_{j|i}\log_2 p_{j|i}$$

σ_i is chosen so that $\mathrm{Perp}(P_i) =$ perplexity for all i.

van der Maaten & Hinton (2008) JMLR · Damrich et al. (2023) Sec. 2

Formal Definitions II: NCE, NEG and the Normalisation Parameter

Noise-Contrastive Estimation (NCE). Turns density estimation into binary classification against noise distribution ξ:

$$\mathcal{L}^{\text{NCE}}(\theta) = -\mathbb{E}_{s\sim p} \log \frac{q_\theta(s)}{q_\theta(s) + m\xi(s)} - m\,\mathbb{E}_{t\sim\xi} \log \frac{m\xi(t)}{q_\theta(t) + m\xi(t)}$$

Minimiser: $q_{\theta^*} = p$ (recovers the true data distribution; Z is learnable).

Negative Sampling (NEG). Same structure, but replaces learnable Z with a *fixed* $\bar{Z}$:

$$\mathcal{L}^{\text{NEG}}(\theta) = -\mathbb{E}_{s\sim p} \log \frac{q_\theta(s)}{q_\theta(s) + 1} - m\,\mathbb{E}_{t\sim\xi} \log \frac{1}{q_\theta(t) + 1}$$

Minimiser: $q_{\theta^*} = (|X|/m)\,p$ — proportional but *not equal* to p.

Generalised NCE (Corollary 2, Damrich et al. 2023). Parametrised by $\bar{Z}$:

$$\mathcal{L}^{\text{NCE}}_{\bar{Z}}(\theta) = -\mathbb{E}_p \log \frac{q_\theta}{q_\theta + \bar{Z}m\xi} - m\,\mathbb{E}_\xi \log \frac{\bar{Z}m\xi}{q_\theta + \bar{Z}m\xi}$$

$\bar{Z} = 1/(m\xi) \Rightarrow$ NCE; $\bar{Z} = |X|/m \Rightarrow$ NEG (UMAP regime); varying $\bar{Z}$ **spans the t-SNE–UMAP spectrum**.

Damrich et al. (2023) arXiv:2206.01816, Corollary 2

Formal Definitions III: Neg-t-SNE and the InfoNCE Connection

Neg-t-SNE (Damrich et al. 2023). Apply NEG to the t-SNE k-NN graph with Cauchy kernel and fixed $\bar{Z}$:

$$\mathcal{L}_{\bar{Z}}^{\text{Neg-t-SNE}} = -\sum_{(i,j)\in\mathcal{E}} w_{ij} \log q_{ij} - m \sum_{(i,j)\notin\mathcal{E}} \log(1 - q_{ij}), \quad q_{ij} = \frac{\tilde{q}_{ij}}{\tilde{q}_{ij} + \bar{Z}}, \quad \tilde{q}_{ij} = (1 + d_{ij}^2)^{-1}$$

Key identities (Damrich et al. 2023, Theorems 1–3):

- $\bar{Z} = Z^{\text{t-SNE}} \implies$ Neg-t-SNE $\equiv$ t-SNE optimum
- $\bar{Z} = Z^{\text{NCVis}} \implies$ Neg-t-SNE $\equiv$ NC-t-SNE (NCVis) optimum
- $\bar{Z} = \binom{n}{2}/m \implies$ Neg-t-SNE $\equiv$ UMAP optimum
- For large n: $\bar{Z}^{\text{UMAP}} \sim n^2/2m \gg Z^{\text{t-SNE}} \sim 50n$

InfoNC-t-SNE. Replace NCE with InfoNCE loss (multinomial NM classification):

$$\mathcal{L}^{\text{InfoNC}} = -\mathbb{E}_{x\sim p} \log \frac{q_\theta(x)}{q_\theta(x) + \sum_{i=1}^{m} q_\theta(x_i)}$$

Result: embeddings similar to NC-t-SNE; directly comparable to SimCLR.

Damrich et al. (2023) arXiv:2206.01816, Sections 4–7

Formal Definitions IV: Evaluation Metrics

k-NN recall. Fraction of true high-D neighbours recovered in the embedding:

$$\mathrm{Recall}(k) = \frac{1}{nk} \sum_{i=1}^{n} |V_i^{(k)} \cap W_i^{(k)}|$$

$V_i^{(k)}$: top-k neighbours in high-D; $W_i^{(k)}$: top-k neighbours in low-D.

Trustworthiness. Penalises intrusions (low-D neighbours that were far in high-D):

$$\mathrm{TW}(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^{n} \sum_{j \in \mathcal{U}_k(i)} (r_{ij} - k)$$

Spearman correlation of pairwise distances. Global structure quality: $\rho(\{d_{ij}^{\mathrm{high}}\}, \{d_{ij}^{\mathrm{low}}\})$.

Multi-scale quality (de Bodt & Lee). $Q_{\mathrm{NX}}(k) = \frac{1}{n} \sum_i \frac{|V_i^{(k)} \cap W_i^{(k)}|}{k}$; area under the $Q_{\mathrm{NX}}(k)$ curve gives a single scalar.

No single metric suffices — always evaluate at multiple scales k.

Damrich et al. (2023) use k-NN recall and pairwise distance correlation to validate Neg-t-SNE

UMAP is NEG Applied to the t-SNE Framework

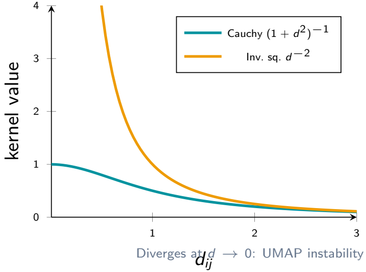
Lemma 3 (Damrich et al. 2023). UMAP's loss is NEG applied with the inverse-square kernel $\tilde{q}_\theta(ij) = 1/d_{ij}^2$:

$$q_\theta(ij) = \frac{1}{1 + d_{ij}^2} = \frac{\tilde{q}_\theta(ij)}{\tilde{q}_\theta(ij) + 1}$$

The Cauchy kernel $(1 + d^2)^{-1}$ and inverse-square d^{-2} agree for large d but diverge as $d \to 0$ — causing **UMAP numerical instability**.

NC-t-SNE (NCVis). Uses NCE to approximate t-SNE; learns normalisation Z so $Z^{\text{NC-t-SNE}} \approx Z^{\text{t-SNE}}$ at convergence. **Same optimum as t-SNE; much faster in practice.**

Damrich et al. (2023) arXiv:2206.01816, Section 6



Diverges at $d \to 0$: UMAP instability

The Neg-t-SNE Spectrum: Interpolating Between t-SNE and UMAP

Neg-t-SNE replaces learnable Z with fixed $\bar{Z}$ in the NCE loss:

- $\bar{Z}$ small $\Rightarrow$ spread embedding (t-SNE-like)
- $\bar{Z}$ large $\Rightarrow$ compact clusters (UMAP-like)

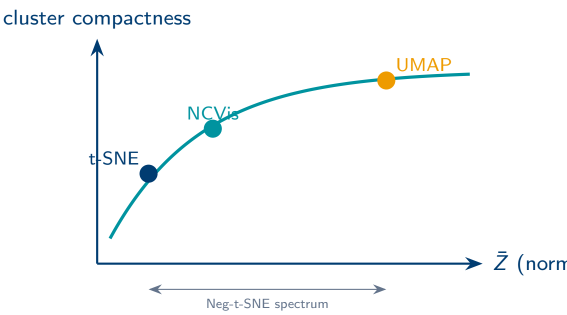
Specific fixed points:

- $\bar{Z} = Z^{\text{t-SNE}}$ $\rightarrow$ reproduces t-SNE
- $\bar{Z} = Z^{\text{NCVis}}$ $\rightarrow$ reproduces NCVis
- $\bar{Z} = \binom{n}{2}/m$ $\rightarrow$ gives UMAP

For large n: $\bar{Z}^{\text{UMAP}} \sim n^2 \gg Z^{\text{t-SNE}} \sim 50n$. This explains why UMAP clusters are more compact.

Practical implication: explore multiple $\bar{Z}$ values to guard against over-interpretation of any single embedding.

Damrich et al. (2023) arXiv:2206.01816, Figure 1 & Section 5



See Fig. 1 of Damrich et al. (2023): MNIST spectrum from $\bar{Z} < Z^{\text{t-SNE}}$ (a) to $\bar{Z} > |X|/m$ (e).

Contrastive NE and Self-Supervised Learning

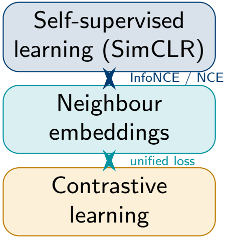
The formal connection (Damrich et al. 2023)

	Neighbour Embedding	SSL (SimCLR)
Positives	k-NN graph edges	Augmented views
Negatives	Random pairs	Other batch items
Loss	NCE / NEG	InfoNCE
Repr. dim	2	$\sim$128
Mapping	Non-parametric	Neural network

Self-supervised representation learning (SimCLR, MoCo) and neighbour embeddings are the **same objective family**.

Research frontiers:

- Data augmentations improve neighbour embeddings?
- Parametric UMAP / t-SNE via neural nets
- Combining LLM encoders with NE



Damrich et al. (2023): same objective family

Table 1: CIFAR-10 Classification — InfoNCE Saturates Early

Test set classification accuracies on CIFAR-10 representations obtained with InfoNCE loss and batch size $b = 1024$.

Accuracy saturates already at a low number of noise samples m, which is **common behaviour for neighbour embeddings**.

Method	$m = 1$	$m = 2$	$m = 4$	$m = 8$	$m = 16$	$m = 64$	$m = 511$
InfoNCE (linear)	82.0	84.1	85.3	86.0	86.3	86.7	86.8
InfoNCE (kNN)	81.4	83.7	85.0	85.8	86.1	86.5	86.6
NCE (linear)	74.3	78.9	82.1	84.2	85.4	86.3	86.7
NCE (kNN)	73.8	78.4	81.7	83.9	85.1	86.1	86.5
NEG (linear)	55.1	62.3	70.4	76.8	81.2	85.1	86.5
NEG (kNN)	54.7	61.8	69.9	76.3	80.7	84.8	86.3

InfoNCE with $m \geq 16$ already matches the $m = 511$ (full-batch) baseline. NEG (used by UMAP) needs more noise samples to reach comparable accuracy.

Damrich et al. (2023) arXiv:2206.01816, Table 1 (adapted)

Attraction–Repulsion Trade-off

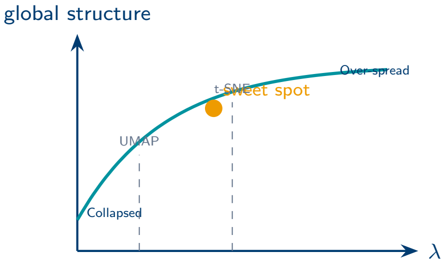
All neighbour embeddings minimise:

$$\mathcal{L} = \underbrace{\sum_{(i,j)\in\mathcal{E}} w_{ij} f_{\mathrm{attr}}(\mathbf{y}_i, \mathbf{y}_j)}_{\text{attractive}} - \lambda \underbrace{\sum_{(i,j)\notin\mathcal{E}} f_{\mathrm{rep}}(\mathbf{y}_i, \mathbf{y}_j)}_{\text{repulsive}}$$

λ = repulsion strength; larger $\bar{Z}$ in Neg-t-SNE $\approx$ stronger attraction.

- Low $\lambda \to$ compact clusters (UMAP-like)
- High $\lambda \to$ more global structure (t-SNE-like)
- UMAP uses negative sampling: $O(n)$ per step

Böhm, Berens & Kobak (2022) JMLR · Damrich et al. (2023) Sec. 5



Key Hyperparameters and Their Effects

t-SNE: `perplexity`

$$\text{perplexity} \approx k_{\text{effective neighbours}}$$

- Small: tight local clusters
- Large: more global layout
- **Best practice:** perplexity $= n/100$ for large n, or **multi-scale** $(30 + n/100)$
- Typical range: 5–500

Neg-t-SNE: $\bar{Z}$

- $\bar{Z} \ll Z^{\text{t-SNE}}$: spread, few distinct clusters
- $\bar{Z} = Z^{\text{t-SNE}}$: t-SNE-like layout
- $\bar{Z} = |X|/m$: UMAP-like compact clusters

Kobak & Linderman (2021) · Damrich et al. (2023) Sec. 6

UMAP: `n_neighbors`, `min_dist`

- `n_neighbors`: neighbourhood size ($\approx$ perplexity)
- `min_dist`: cluster compactness
- `metric`: encodes domain knowledge (cosine, correlation, ...)
- **Best practice:** `n_neighbors` 15–50; `min_dist` 0.1–0.5

Critical: PCA initialisation. Always initialise t-SNE and UMAP from PCA (not random). Damrich et al. (2023) confirm it is essential for stability in Neg-t-SNE. Benefits: (i) reproducibility, (ii) global structure seeded correctly, (iii) faster convergence.

Notebook Playground: github.com/berenslab/contrastive-ne

The official code repository for Damrich et al. (2023) provides Jupyter notebooks to reproduce all figures and experiment interactively.

Getting started:

```
git clone https://github.com/berenslab/
     contrastive-ne
cd contrastive-ne
pip install -e .
jupyter notebook
```

Key notebooks:

- negsne_spectrum.ipynb — vary $\bar{Z}$ on MNIST
- infonce_tsne.ipynb — InfoNC-t-SNE vs SimCLR
- ncvis_comparison.ipynb — NCVis / NC-t-SNE

Core API (from the repo):

```
from contrastive_ne import NegTSNE

# Run Neg-t-SNE with varying Zbar
model = NegTSNE(
    n_components=2,
    Zbar=1e8,       # try: 1e6, 1e7, 1e8, 1
        e9
    perplexity=30,
    init='pca',     # always use PCA init
    random_state=42
)
Y = model.fit_transform(X_pca)
```

Sweep $\bar{Z}$ values:

```
import numpy as np
Zbars = np.logspace(6, 10, 9)
for Zbar in Zbars:
    Y = NegTSNE(Zbar=Zbar,
                init='pca').fit_transform(
                    X)
    # compare kNN recall across Zbars
```

Lab assignment: Use the playground to run the $\bar{Z}$

The Local–Global Trade-off

What neighbour embeddings **preserve**:

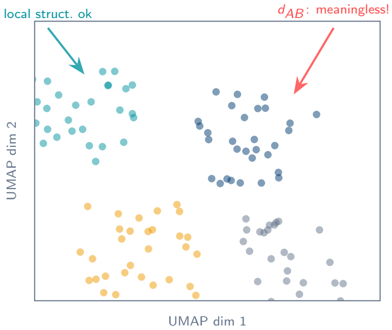
- ✓ Local k-NN neighbourhood structure
- ✓ Within-cluster geometry
- ✓ Rough cluster separation

What they **distort or destroy**:

- × **Between-cluster distances** are meaningless
- × **Cluster sizes** do not reflect density
- × **Topological features** (loops, voids) often lost
- × **Continuous gradients** can be torn

Global methods are better for:

- Isomap, PHATE: geodesic distances on manifold
- Laplacian Eigenmaps: spectral structure
- Use 5–50 dims for downstream HDBSCAN clustering



Common Pitfalls and How to Diagnose Them

Pitfall 1: Crowding artefact
High-D manifold points pile up at the centre.
Diagnosis: dense blob, no structure.
Fix: increase perplexity; early exaggeration.

Pitfall 2: Torn trajectories
Continuous progressions fragmented into separate clusters.
Diagnosis: known progression $\rightarrow$ disjoint blobs.
Fix: use PHATE, larger perplexity, or increase $\bar{Z}$.

Pitfall 3: UMAP instability
Fuzzy clusters from kernel divergence at $d_{ij} \rightarrow 0$.
Fix: learning-rate annealing; use Neg-t-SNE with Cauchy kernel.

Pitfall 4: Hyperparameter sensitivity
Different perplexity / `n_neighbors` $\rightarrow$ different layouts.
Fix: always show ≥ 3 settings.

Pitfall 5: Random init instability
Different seeds $\rightarrow$ mirror images / different orientations.
Fix: PCA init + fixed seed.

The Embedding Multiverse
Vary $\bar{Z}$ (Damrich et al.) *and* hyperparameters (Coupette et al.) — map the **distribution** of embeddings, not a single run.

Summary

Key ideas from today (Damrich et al. 2023)

1. t-SNE uses MLE; UMAP uses NEG with fixed $\bar{Z}$
2. NCE and NEG are related by fixed vs. learned normalisation Z
3. UMAP and t-SNE are **instances of the same contrastive family**
4. Varying $\bar{Z}$ in Neg-t-SNE spans the full spectrum
5. UMAP's compact clusters: $\bar{Z}^{\text{UMAP}} \sim n^2 \gg Z^{\text{t-SNE}} \sim 50n$
6. UMAP instability $\leftarrow$ inverse-square kernel diverges at $d \to 0$
7. Neighbour embeddings $\equiv$ contrastive self-supervised learning
8. **PCA init** is essential: always use it

Readings

- **Primary:** Damrich et al. (2023) `arXiv:2206.01816` — all sections
- Kobak & Berens (2019) *Nature Comms*
- Böhm et al. (2022) JMLR — attraction-repulsion
- Wattenberg et al. (2016) Distill — how to use t-SNE

Lab assignment

- Clone `berenslab/contrastive-ne`
- Run the $\bar{Z}$ spectrum on MNIST; reproduce Fig. 1
- Compare t-SNE, Neg-t-SNE, UMAP on MNIST
- Compute k-NN recall at $k \in \{10, 50, 100\}$

Thank you

Questions?

Damrich, Böhm, Hamprecht & Kobak (2023) `arXiv:2206.01816` ICLR

Code: `github.com/berenslab/contrastive-ne`

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